

Smooth-point certificates for Polydegree containments

A Jacobian interpretation of the Lewis–Perry–Straub determinant

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Abstract

Lewis, Perry and Straub introduced integral coefficient polynomials $g_{n,e}$ and an auxiliary determinant $a_{d,e}$ whose simultaneous specialisation yields containments between strata of plane polynomial automorphisms. We make explicit a differential structure hidden in their definitions. The entries of the determinant are precisely negative partial derivatives,

$$\alpha_{i,j,e} = -\partial_{x_{i+1}} g_{j+1,e},$$

so $a_{d,e}$ is, up to the sign $(-1)^e$, the Jacobian determinant of $g_{d,e}, \dots, g_{d+e-1,e}$. Weighted Euler identities then give an exact integral syzygy. In the chart $x_1 = 1$, this converts the original non-vanishing condition into a smooth-point condition for $g_{d,e} = \dots = g_{d+e-2,e} = 0$ together with avoidance of the next coefficient hypersurface.

This reformulation yields a smaller finite-field certificate. A simple common zero modulo a prime with non-zero next coefficient lifts to characteristic zero; no condition excluding primes dividing $d + e - 1$ is needed. We also prove a curve-theoretic bridge in which a reduced affine zero divisor supplies the required Jacobian point. Exact computations for $e = 3$ show that, for $2 \leq d \leq 20$, the entire affine complete intersection is reduced and avoids the next coefficient. We also prove that every one-variable boundary factor has only simple positive roots, while adjacent boundary coprimality is checked through $d = 200$. These results motivate, but do not prove, a uniform all- d affine conjecture. The paper separates proved identities, inherited theorems, exact finite evidence and open claims throughout.

Keywords: plane polynomial automorphisms; Polydegree Conjecture; Jacobian criterion; multivariate Hensel lifting; weighted homogeneous polynomials; complete intersections; computer-assisted mathematics

1 Introduction

The group of polynomial automorphisms of the complex affine plane carries two structures whose interaction is subtle. The Jung–van der Kulk decomposition assigns a polydegree sequence to each automorphism, while the group is also an ind-variety in which one can ask when one polydegree stratum lies in the Zariski closure of another. Lewis, Perry and Straub recast a family of these containment questions in terms of explicit coefficient polynomials and proved a specialisation theorem: suitable complex values of $x_1, \dots, x_e$ imply

$$\mathcal{G}_{(d+e)} \subseteq \overline{\mathcal{G}_{(d,e+1)}}. \quad (1.1)$$

Their formulation makes the problem algorithmically decidable for fixed (d, e) and produced computer-aided ranges for $e = 3, 4, 5$ (Lewis et al. 2019). Perry’s earlier thesis developed the same computational direction for the case later indexed by Lewis–Perry–Straub as $e = 3$ (Perry 2016). The coefficient polynomials also sit within the rigidity and Strong Factorial framework of Edo and van den Essen (Edo and Essen 2014).

The determinant appearing in the Lewis–Perry–Straub criterion was described there as presently unmotivated (Lewis et al. 2019). The starting point of this paper is that it is not an additional object: it is the Jacobian determinant of the coefficient map already under study. Once this observation is combined with weighted homogeneity, the original specialisation criterion becomes a local geometric question:

Find one smooth point of the partial coefficient complete intersection that does not lie on the next coefficient hypersurface.

That change of language has concrete consequences. It removes the full $e \times e$ determinant from a finite-field search, permits useful primes previously rejected by a residue-level determinant test, and joins two seemingly different proof routes: Hensel lifting from a finite field and divisor separation on a curve.

1.1 Contributions

The proved contributions are as follows.

1. We give a formal-inverse proof that every displayed rational multinomial formula for $g_{n,e}$ is an element of $\mathbf{Z}[x_1, \dots, x_e]$.
2. We prove the derivative and determinant identities

$$\alpha_{i,j,e} = -\frac{\partial g_{j+1,e}}{\partial x_{i+1}}, \quad a_{d,e} = (-1)^e \det D(g_{d,e}, \dots, g_{d+e-1,e}).$$

3. We derive an exact weighted-Euler syzygy, not merely an equality after evaluation on a zero locus. It identifies the Lewis–Perry–Straub non-vanishing condition with the non-vanishing of an affine Jacobian minor and the next coefficient.
4. We prove a reduced finite-field certificate theorem. A modular common zero at which the affine minor and the next coefficient are non-zero suffices. The prime may divide $d + e - 1$: the original determinant then vanishes in the residue field but remains non-zero at the characteristic zero lift.
5. We prove a prime–gap–Jacobian bridge on a smooth complete-intersection curve, including the affine-support and perfect-ground-field hypotheses needed for coefficient-one prime-divisor applications.
6. For $e = 3$ we prove uniform squarefreeness of the one-variable boundary factors by a terminating hypergeometric factorisation and finite free convolution. We also report exact finite computations that test a stronger scheme-theoretic conjecture. For $2 \leq d \leq 20$, every affine common point is smooth and avoids the next coefficient, and adjacent boundary coprimality is checked through $d = 200$. Neither bounded sweep is promoted to an all- d theorem.

1.2 Relation to prior results and novelty boundary

Lewis–Perry–Straub Theorem 13 records the ranges

$$e = 3, \ 2 \leq d < 50; \quad e = 4, \ 2 \leq d < 20; \quad e = 5, \ 2 \leq d \leq 12,$$

while their Theorem 14, attributed to Edo, proves (1.1) whenever $d - 1$ is divisible by e (Lewis et al. 2019). In particular, for $e = 3$ the class $d \equiv 1 \pmod{3}$ is already an infinite family. The present paper therefore does not claim the first infinite family, nor does it claim an all- d result.

A bounded search of the primary paper, Perry’s thesis, the Strong Factorial paper and indexed citing literature did not locate the derivative–Jacobian packaging developed here. That is provisional negative evidence, not proof of priority. The safe claim is that we **make explicit and exploit** a Jacobian structure that was not found in the sources inspected through 9 August 2026. Authenticated specialist-database searches, original-author contact, expert review and journal peer review remain outstanding.

1.3 Evidence classes

Four evidence classes are used below.

Label	Meaning
Proved	A general argument is included in this manuscript.
Inherited	A result is used from a cited external source, chiefly Lewis–Perry–Straub Theorem 4.
Exactly checked	A finite statement was evaluated with exact arithmetic and retained receipts.
Open	A conjecture or priority question is explicitly unresolved.

Internal replay, code-path diversity and mutation controls improve confidence in the exact computations. They are not independent reproduction, formal verification, expert acceptance or peer review.

2 Coefficient polynomials and formal inversion

Fix $e \geq 1$. For $\mathbf{a} = (a_1, \dots, a_e) \in \mathbf{Z}_{\geq 0}^e$, write

$$|\mathbf{a}| = a_1 + \dots + a_e, \quad \mathbf{a} \cdot \mathbf{N} = a_1 + 2a_2 + \dots + ea_e.$$

For $n \geq 0$, define

$$g_{n,e} = \frac{1}{n+1} \sum_{\mathbf{a} \cdot \mathbf{N} = n} (-1)^{|\mathbf{a}|} \binom{n+|\mathbf{a}|}{\mathbf{a}, n} \mathbf{x}^{\mathbf{a}}. \quad (2.1)$$

For $0 \leq i \leq e-1$ and $j \geq -1$, define

$$\alpha_{i,j,e} = \sum_{\mathbf{a} \cdot \mathbf{N} = j-i} (-1)^{|\mathbf{a}|} \binom{|\mathbf{a}|+j+2}{\mathbf{a}, j+2} \mathbf{x}^{\mathbf{a}}, \quad (2.2)$$

with an empty sum when $j-i < 0$, and put

$$a_{d,e} = \det(\alpha_{i,j,e})_{0 \leq i \leq e-1, d-1 \leq j \leq d+e-2}. \quad (2.3)$$

These are the definitions used by Lewis, Perry and Straub (Lewis et al. 2019). The factor $1/(n+1)$ in (2.1) makes integrality worth stating before reduction modulo a prime.

Lemma 2.1 (integrality and inverse coefficients) [Proved]

Let $I(w)$ be the unique compositional inverse of

$$H(t) = t + x_1 t^2 + \dots + x_e t^{e+1}.$$

Then

$$I(w) \in \mathbf{Z}[x_1, \dots, x_e][[w]], \quad g_{n,e} = [w^{n+1}]I(w). \quad (2.4)$$

Proof

Write $I(w) = w + \sum_{m \geq 2} c_m w^m$. In the identity

$$I + x_1 I^2 + \cdots + x_e I^{e+1} = w,$$

the coefficient of w^m is c_m plus an integer polynomial in $x_1, \dots, x_e, c_2, \dots, c_{m-1}$. Induction therefore gives $c_m \in \mathbf{Z}[x_1, \dots, x_e]$.

Formal Lagrange inversion gives

$$[w^{n+1}]I(w) = \frac{1}{n+1} [t^n] (1 + x_1 t + \cdots + x_e t^e)^{-(n+1)}.$$

Expanding the negative power multinomially yields exactly (2.1). Hence the apparently rational expression is an integral polynomial. $\square$

The recursion in the first paragraph and the multinomial expression in the second are algebraically different constructions of the same coefficients. They are used as separate internal replay paths in the evidence package.

3 The auxiliary determinant is a Jacobian

Proposition 3.1 (derivative identity) [Proved]

For $0 \leq i \leq e-1$ and $j \geq -1$,

$$\boxed{\alpha_{i,j,e} = -\frac{\partial g_{j+1,e}}{\partial x_{i+1}}.} \tag{3.1}$$

Proof

Only terms of $g_{j+1,e}$ whose x_{i+1} exponent is positive survive differentiation. Put $\mathbf{b} = \mathbf{a} + \mathbf{e}_{i+1}$. Then

$$\mathbf{a} \cdot \mathbf{N} = j - i, \quad |\mathbf{b}| = |\mathbf{a}| + 1.$$

After multiplication by $b_{i+1} = a_{i+1} + 1$, the coefficient of $\mathbf{x}^{\mathbf{a}}$ in the derivative is

$$\begin{aligned} & \frac{1}{j+2} (-1)^{|\mathbf{a}|+1} \frac{(j+1+|\mathbf{b}|)!}{b_1! \cdots b_e! (j+1)!} (a_{i+1} + 1) \\ &= (-1)^{|\mathbf{a}|+1} \frac{(j+2+|\mathbf{a}|)!}{a_1! \cdots a_e! (j+2)!}. \end{aligned}$$

This is the negative of the coefficient in (2.2). Summing over the common index set proves (3.1). $\square$

For $d \geq 2$ and $e \geq 2$, let

$$D_{d,e} = \left(\frac{\partial g_{d+r,e}}{\partial x_s} \right)_{0 \leq r \leq e-1, 1 \leq s \leq e}.$$

Corollary 3.2 (full determinant identity) [Proved]

$$\boxed{a_{d,e} = (-1)^e \det D_{d,e}.} \quad (3.2)$$

Proof

The matrix in (2.3) is $-D_{d,e}^\top$ by Proposition 3.1. Its determinant is therefore $(-1)^e \det D_{d,e}$. $\square$

Thus $a_{d,e}$ tests whether the full coefficient map

$$(x_1, \dots, x_e) \mapsto (g_{d,e}, \dots, g_{d+e-1,e})$$

is locally invertible. The specialisation theorem does not, however, require all e coefficients to vanish. The next section extracts the relevant minor on the partial zero locus.

4 Weighted Euler syzygy and the smooth-point criterion

Give x_s weight s . Each $g_{n,e}$ is weighted homogeneous of weight n , so

$$\sum_{s=1}^e s x_s \frac{\partial g_{n,e}}{\partial x_s} = n g_{n,e}. \quad (4.1)$$

Define the affine minor

$$J_{d,e} = \det \left(\frac{\partial g_{d+r,e}}{\partial x_s} \right)_{0 \leq r \leq e-2, 2 \leq s \leq e}. \quad (4.2)$$

For $0 \leq r \leq e-1$, let M_r be the determinant of the $(e-1) \times (e-1)$ submatrix of $D_{d,e}$ obtained by deleting row r and the x_1 column. In particular, $M_{e-1} = J_{d,e}$.

Proposition 4.1 (exact weighted-Euler syzygy) [Proved]

In $\mathbf{Z}[x_1, \dots, x_e]$,

$$\boxed{x_1 a_{d,e} + (d+e-1) g_{d+e-1,e} J_{d,e} = \sum_{r=0}^{e-2} (-1)^{e+r} (d+r) g_{d+r,e} M_r.} \quad (4.3)$$

Consequently,

$$x_1 a_{d,e} \equiv -(d+e-1) g_{d+e-1,e} J_{d,e} \pmod{(g_{d,e}, \dots, g_{d+e-2,e})}. \quad (4.4)$$

Proof

In $D_{d,e}$ replace the first column by

$$C = \sum_{s=1}^e s x_s (D_{d,e})_s.$$

By multilinearity, the resulting determinant is $x_1 \det D_{d,e}$: every summand with $s \geq 2$ repeats an existing column. Equation (4.1) makes entry r of C equal to $(d+r)g_{d+r,e}$. Expanding down the first column and using (3.2) gives

$$x_1 a_{d,e} = \sum_{r=0}^{e-1} (-1)^{e+r} (d+r) g_{d+r,e} M_r.$$

The last summand is $-(d+e-1)g_{d+e-1,e}J_{d,e}$. Moving it to the left proves (4.3), and (4.4) follows. $\square$

Corollary 4.2 (scheme-theoretic smooth-point reformulation) [Proved]

Let K be a field and let $P \in K^e$ satisfy

$$g_{d,e}(P) = \cdots = g_{d+e-2,e}(P) = 0.$$

Assume $x_1(P) \neq 0$ and $\text{char } K \nmid d+e-1$. Then the following are equivalent:

1. $g_{d+e-1,e}(P) \neq 0$ and $a_{d,e}(P) \neq 0$;
2. $g_{d+e-1,e}(P) \neq 0$ and $J_{d,e}(P) \neq 0$;
3. P lies outside $V(g_{d+e-1,e})$, and the closed subscheme

$$\text{Spec } K[x_1, \dots, x_e] / (g_{d,e}, \dots, g_{d+e-2,e})$$

is smooth of codimension $e-1$ at P .

Proof

On the common zero locus, (4.4) makes (1) and (2) equivalent under the stated unit hypotheses. For each of the first $e-1$ coefficient polynomials, (4.1) expresses its x_1 derivative as a linear combination of its $x_2, \dots, x_e$ derivatives because $x_1(P) \neq 0$. Hence the partial Jacobian has rank $e-1$ exactly when the square minor (4.2) is non-zero. The Jacobian criterion for a subscheme cut out by $e-1$ equations in affine e -space gives the equivalence with (3). $\square$

The characteristic restriction belongs to this residue-field equivalence. It will disappear from the finite-field-to-characteristic-zero theorem.

5 A smaller Hensel certificate

Work in the chart $x_1 = 1$ and put

$$F_r(x_2, \dots, x_e) = g_{d+r,e}(1, x_2, \dots, x_e). \quad (5.1)$$

The system $F_0 = \dots = F_{e-2} = 0$ has $e - 1$ equations in $e - 1$ variables, and its Jacobian determinant is $J_{d,e}(1, x_2, \dots, x_e)$.

Theorem 5.1 (smooth finite-field certificate) [Proved]

Let $d \geq 2$ and $e \geq 2$. Suppose that a prime p and a point $Q \in \mathbf{F}_p^{e-1}$ satisfy

$$F_0(Q) = \dots = F_{e-2}(Q) = 0, \quad F_{e-1}(Q)J_{d,e}(1, Q) \neq 0. \quad (5.2)$$

Then there is a complex specialisation satisfying the hypotheses of Lewis–Perry–Straub Theorem 4. Consequently,

$$\mathcal{G}_{(d+e)} \subseteq \overline{\mathcal{G}_{(d,e+1)}}. \quad (5.3)$$

Proof

The non-zero Jacobian determinant makes Q a simple zero of the square system $F_0 = \dots = F_{e-2} = 0$. Lemma 2.1 shows that all these polynomials have integral coefficients. Multivariate Hensel lifting therefore produces an exact lift $\tilde{Q} \in \mathbf{Z}_p^{e-1}$ in the same residue class. The values $F_{e-1}(\tilde{Q})$ and $J_{d,e}(1, \tilde{Q})$ remain p -adic units. This is precisely the simple-root setting of multivariate Hensel lifting (Grochow 2025).

Consider the localisation

$$R = \frac{\mathbf{Q}[x_2, \dots, x_e, 1/(F_{e-1}J_{d,e})]}{(F_0, \dots, F_{e-2})}, \quad (5.4)$$

where $J_{d,e}$ is evaluated at $x_1 = 1$. Evaluation at $\tilde{Q}$ defines a unital $\mathbf{Q}$ -algebra homomorphism $R \rightarrow \mathbf{Q}_p$, so $R \neq 0$. The extension $\mathbf{Q} \subset \mathbf{C}$ is faithfully flat; hence $R \otimes_{\mathbf{Q}} \mathbf{C} \neq 0$ (Stacks project authors 2026a). This is a non-zero finite-type $\mathbf{C}$ -algebra. A maximal ideal and the weak Nullstellensatz give a complex point at which the first $e - 1$ equations vanish and $F_{e-1}J_{d,e}$ does not (Stacks project authors 2026b). Corollary 4.2, now in characteristic zero, shows that $a_{d,e}$ is non-zero there. The inherited Lewis–Perry–Straub Theorem 4 yields (5.3) (Lewis et al. 2019). $\square$

The proof does **not** assert an embedding $\mathbf{Q}_p \hookrightarrow \mathbf{C}$. The bridge is the non-zero localised $\mathbf{Q}$ -algebra and faithful scalar extension.

Remark 5.2 (bad-characteristic rescue)

No assumption $p \nmid d + e - 1$ is needed in Theorem 5.1. If $p \mid d + e - 1$, equation (4.4) forces the residue of $a_{d,e}$ to vanish at a finite-field common zero. At the exact p -adic lift, however,

$$a_{d,e}(1, \tilde{Q}) = -(d + e - 1)F_{e-1}(\tilde{Q})J_{d,e}(1, \tilde{Q}) \neq 0 \quad \text{in } \mathbf{Q}_p. \quad (5.5)$$

The integer $d + e - 1$ may be a nonunit in $\mathbf{Z}_p$, but it is not zero in the characteristic-zero field $\mathbf{Q}_p$. Thus the old modular $a_{d,e}$ -unit test is strictly stronger than the certificate actually needed for complex existence.

Example 5.3 (exact positive regression) [Exactly checked]

For $e = 3$, $d = 8$, $p = 5$ and $(x_1, x_2, x_3) = (1, 3, 0)$, exact finite-field evaluation gives

$$g_{8,3} = g_{9,3} = 0, \quad g_{10,3} = 2, \quad J_{8,3} = 3, \quad a_{8,3} = 0 \pmod{5}.$$

The legacy determinant-unit certificate rejects this point. Theorem 5.1 accepts it, and (5.5) proves that the determinant at the exact 5-adic lift is non-zero. The fixture is retained as a positive regression test; it is not merely an illustrative calculation.

6 A curve-theoretic route to the same point

The smooth-point criterion also clarifies what a divisor argument must prove. The statement below deliberately distinguishes an algebraically closed geometric theorem from descent over a perfect ground field.

Theorem 6.1 (prime-gap-Jacobian bridge) [Proved]

Work over $\mathbf{C}$ in the chart $x_1 = 1$. Suppose the first $e - 2$ polynomials

$$F_0, \dots, F_{e-3}$$

cut out a smooth, geometrically integral complete-intersection curve $C \subset \mathbf{A}^{e-1}$ **scheme-theoretically**. Let $\overline{C}$ be its smooth projective completion, and regard the F_r as rational functions on $\overline{C}$. Assume:

1. $F_{e-2}|_{\overline{C}}$ is non-zero, and its zero divisor is non-empty, reduced and supported inside the affine open C ;
2. $F_{e-1}(Q) \neq 0$ at every point $Q \in \text{Supp}(F_{e-2}|_{\overline{C}})_0$.

Then every point Q of that zero divisor satisfies

$$F_0(Q) = \dots = F_{e-2}(Q) = 0, \quad F_{e-1}(Q)J_{d,e}(1, Q) \neq 0. \quad (6.1)$$

Consequently, each such point supplies a Lewis–Perry–Straub specialisation.

Proof

The support condition makes Q an affine point at which all displayed functions are regular. Smooth complete-intersection status says that $dF_0, \dots, dF_{e-3}$ are independent and cut the tangent space down to the one-dimensional space $T_Q C$. Reducedness of the zero divisor means that $F_{e-2}|_C$ has a simple zero, equivalently $dF_{e-2}|_{T_Q C} \neq 0$. Hence $dF_0, \dots, dF_{e-2}$ are independent. Their square Jacobian determinant is $J_{d,e}(1, Q)$, so it is non-zero. The second hypothesis supplies the other factor in (6.1), and Corollary 4.2 gives the original determinant condition. $\square$

The requirement that C be the scheme-theoretic complete intersection is load-bearing. Smoothness of a selected component would not exclude another component or embedded structure from changing the Jacobian statement. The affine-support condition is equally necessary: a zero at the projective boundary does not define a point in the chart used by Theorem 5.1.

Corollary 6.2 (perfect-ground-field descent) [Proved]

Let $k \subset \mathbf{C}$ be a perfect field of characteristic zero, and choose its algebraic closure $\bar{k}$ inside $\mathbf{C}$. Suppose the scheme-theoretic complete intersection C , its smooth projective completion and the F_r are defined over k . Assume the zero divisor of F_{e-2} is one closed point with coefficient one, is supported in the affine open C , and is disjoint from the zero and pole support of F_{e-1} . Then, after base change to $\bar{k}$, every geometric point above that closed point satisfies Theorem 6.1.

Proof

Over a perfect field, the residue extension of a closed point is separable. A coefficient-one closed point therefore becomes a disjoint reduced collection of geometric points after base change. Affine support and disjointness from the zero and pole support of F_{e-1} are preserved by base change, so Theorem 6.1 applies. $\square$

Over an algebraically closed field, an individual closed point has degree one. A degree- h coefficient-one prime certificate must therefore be formulated over its ground field and then base-changed; phrasing it only over $\mathbf{C}$ loses the arithmetic degree information.

7 Exact $e = 3$ geometry and the uniform conjecture

Set

$$G_n(X, Y) = g_{n,3}(1, X, Y), \quad I_d = (G_d, G_{d+1}) \subset \mathbf{Q}[X, Y],$$

and

$$J_d = \det \frac{\partial(G_d, G_{d+1})}{\partial(X, Y)}.$$

The following section reports exact finite calculations. It is not used to prove the general identities above.

Proposition 7.1 (verified affine range) [Exactly checked]

For every $2 \leq d \leq 20$, exact Gr"obner-basis computation over $\mathbf{Q}$ establishes:

1. $g_{d,3}$ and $g_{d+1,3}$ have homogeneous greatest common divisor 1;
2. I_d is zero-dimensional and

$$\dim_{\mathbf{Q}} \mathbf{Q}[X, Y]/I_d = \left\lfloor \frac{d(d+1)}{6} \right\rfloor; \tag{7.1}$$

3. $I_d + (J_d) = (1)$;
4. $I_d + (G_{d+2}) = (1)$.

The quotient lengths for $d = 2, \dots, 20$ are

$$1, 2, 3, 5, 7, 9, 12, 15, 18, 22, 26, 30, 35, 40, 45, 51, 57, 63, 70.$$

Because I_d is a height-two complete intersection in the smooth surface $\mathbf{A}^2$, the third identity says that every geometric point has full Jacobian rank. The entire affine scheme is therefore reduced, indeed finite étale over $\mathbf{Q}$. The fourth identity says scheme-theoretically that no affine common point lies on $G_{d+2} = 0$. Thus every affine point in the verified range is a valid smooth specialisation.

7.1 Newton polygons and the observed length

The Newton polygon of G_n is

$$\Delta_n = \text{conv}\{(b, c) \in \mathbf{Z}_{\geq 0}^2 : 2b + 3c \leq n\}.$$

An exact six-residue-class calculation gives

$$\text{MV}(\Delta_d, \Delta_{d+1}) = \left\lfloor \frac{d(d+1)}{6} \right\rfloor. \quad (7.2)$$

The observed quotient length therefore reaches the Newton mixed-area count. This suggests a toric non-degeneracy proof, but the specific coefficients of G_d, G_{d+1} must be controlled; generic support geometry alone is not a proof for this pair.

7.2 Boundary reduction

Put

$$B_n(X, Y) = g_{n,3}(0, X, Y).$$

Let $\epsilon = n \bmod 2$, $A = (n - 3\epsilon)/2$, and $z = Y^2/X^3$. On the boundary torus $XY \neq 0$,

$$B_n = X^A Y^\epsilon P_n(z),$$

where

$$P_n(z) = \frac{1}{n+1} \sum_{k=0}^{\lfloor A/3 \rfloor} (-1)^{A+\epsilon-k} \frac{(n+A+\epsilon-k)!}{n!(A-3k)!(\epsilon+2k)!} z^k. \quad (7.3)$$

The adjacent coefficient ratio is

$$\frac{[z^{k+1}]P_n}{[z^k]P_n} = -\frac{(A-3k)(A-3k-1)(A-3k-2)}{(n+A+\epsilon-k)(\epsilon+2k+1)(\epsilon+2k+2)}. \quad (7.4)$$

The minimum exponents give a uniform elementary statement: the common monomial factor of B_d and B_{d+1} is

$$X \quad \text{if } d \equiv 1 \pmod{3}, \quad 1 \quad \text{otherwise.} \quad (7.5)$$

No power X^2 and no Y divides both. Ruling out a further torus factor is separate.

Theorem 7.2 (uniform boundary squarefreeness) [Proved]

For every $n \geq 2$, every zero of P_n is real, positive and simple. In particular, P_n is squarefree over every characteristic-zero field.

Proof

Put $M = \lfloor A/3 \rfloor$, write $A = 3M + \delta$ with $\delta \in \{0, 1, 2\}$, and let $c_{n,M}$ be the non-zero leading coefficient of P_n . Set

$$N = n + A + \epsilon, \quad H = N - M + 1, \quad \sigma = \begin{cases} 1/2, & \epsilon = 0, \\ -1/2, & \epsilon = 1, \end{cases} \quad a = -M + \sigma,$$

and

$$(\beta_1, \beta_2) = \begin{cases} (1/3, 2/3), & \delta = 0, \\ (2/3, 4/3), & \delta = 1, \\ (4/3, 5/3), & \delta = 2. \end{cases}$$

Exact division of consecutive coefficients in (7.3) gives the reciprocal identity

$$\frac{s^M P_n(1/s)}{c_{n,M}} = {}_3F_2 \left(\begin{matrix} -M, a, H \\ \beta_1, \beta_2 \end{matrix}; -\frac{4s}{27} \right). \quad (7.6)$$

Let

$$F_n(x) = {}_3F_2 \left(\begin{matrix} -M, a, H \\ \beta_1, \beta_2 \end{matrix}; x \right).$$

The hypergeometric concatenation theorem for degree- M multiplicative finite free convolution gives, up to a non-zero scalar,

$$F_n = p_n \boxtimes_M q_n, \quad p_n = {}_2F_1(-M, a; \beta_1; x), \quad q_n = {}_2F_1(-M, H; \beta_2; x) \quad (7.7)$$

(Martínez-Finkelshtein et al. 2024b). The Pfaff transformation (DLMF 15.8.1), the Jacobi representation (DLMF 18.5.7), and the strict ordering of Jacobi zeros (DLMF 18.16.1) show that p_n has M simple negative roots and q_n has M simple roots in $(0, 1)$. The relevant Jacobi parameter pairs are

$$(\beta_1 - 1, -\sigma), \quad (\beta_2 - 1, H - M - \beta_2),$$

and every entry exceeds -1 (Olver et al. 2026). Sign preservation for finite free convolution therefore makes F_n negative-rooted.

It remains to exclude repeated roots. Reflect $\widehat{p}_n(x) = p_n(-x)$. For $M \geq 2$ its positive roots are simple, so its logarithmic mesh is greater than one. Proposition 2.17 of Martínez-Finkelshtein, Morales and Perales gives

$$\text{lmesh}(\widehat{p}_n \boxtimes_M q_n) \geq \text{lmesh}(\widehat{p}_n) > 1$$

(Martínez-Finkelshtein et al. 2024a). Hence the reflected convolution has simple positive roots; the reflection identity makes F_n simple and negative-rooted. The cases $M = 0, 1$ are respectively constant and linear. Finally, the scale in (7.6) and reciprocation turn the roots into simple positive roots of P_n . $\square$

The logarithmic-mesh step is load-bearing. A broader strict-interlacing wording for a merely non-negative-rooted convolution factor admits a degenerate counterexample; that formulation is not used here.

Proposition 7.3 (verified adjacent boundary range) [Exactly checked]

For every $2 \leq d \leq 200$,

$$\gcd(P_d, P_{d+1}) = 1,$$

so through this range the full boundary gcd is exactly the factor in (7.5).

When $d \equiv 1 \pmod{3}$, the boundary support is the coarse point $[0 : 0 : 1]$. On the μ_3 uniformising cover of the weighted stack $\mathcal{P}(1, 2, 3)$, the lowest terms of the two equations are independent linear coordinates. The intersection is transverse on that cover and contributes $1/3$ to the stack intersection number. This explains the floor in (7.1); $1/3$ is an orbifold intersection contribution, not a fractional coarse-scheme length.

Conjecture 7.4 (uniform affine smoothness) [Open]

For every $d \geq 2$, the ideal (G_d, G_{d+1}) is radical of length

$$\left\lfloor \frac{d(d+1)}{6} \right\rfloor,$$

and both J_d and G_{d+2} are units in its coordinate algebra.

This would prove more than existence: every affine common point would be a valid smooth specialisation. The exact ranges above strongly constrain a future proof, but extending those ranges alone would add little mathematics. The open structural tasks are uniform adjacent boundary coprimality, toric non-degeneracy or radicality, Jacobian non-vanishing, and exclusion of a triple common zero.

8 Computational assurance and reproducibility

8.1 Frozen finite-field corpus

The upstream v0.3.0 archive contains one finite-field witness for each $50 \leq d \leq 100$. Its witness JSON is byte-identical to the earlier v0.2.0 corpus audited in this project. A fresh reconciliation on 9 August 2026 established:

Gate	Result
v0.3.0 manifest	38/38 files match
upstream dual-path verifier	51/51 under ordinary Python
upstream dual-path verifier	51/51 under <code>python -O</code>
independent standard-library symbolic gates	10/10 in both modes
independent witness relations	51/51 in both modes

Gate	Result
independent reports	byte-identical across modes

The independent verifier does not import the archive code. It uses sparse integer dictionaries, a generic permutation determinant and separate weighted-composition enumeration. This is internal code-path diversity, not reproduction by an unaffiliated group.

8.2 Negative controls

Sixteen mutation or negative controls and the positive bad-characteristic fixture of Example 5.3 were exercised under ordinary and optimised Python. They cover wrong derivative and determinant signs, shifted indices, ordinary in place of weighted Euler coefficients, corrupt residues and points, composite moduli, missing rows, singular points, malformed types and an injected non-integral coefficient. All expected outcomes were observed in both modes.

8.3 A retained fail-open defect

The upstream verifier still contains a proof-critical Python `assert` guarding coefficient divisibility before floor division. Optimised Python removes that guard. An injected non-integral numerator is rejected in ordinary mode but silently floor-divided under `python -O`. The unmodified witness corpus passes in both modes, and the independent verifier uses explicit exceptions; there is no evidence that a current row is false. The defect is nevertheless retained in the record because it is a real failure-detection weakness in the upstream release.

8.4 Geometry receipts

The affine sweep uses exact arithmetic over $\mathbf{Q}$ with two independent constructions of each G_n and Singular for ideal computations. The boundary sweep uses exact univariate arithmetic. Ordinary and optimised Python runs produced byte-identical JSON receipts. The evidence archive records software versions, commands and SHA-256 hashes.

8.5 Assurance boundary

The package establishes that the stated scripts replay the stated finite relations and that the written identities survived adversarial sign and index checks. It does not establish novelty, priority, independent reproduction, proof-assistant verification, specialist acceptance, journal acceptance or the open all- d conjecture.

9 Discussion

9.1 What the Jacobian reformulation changes

The determinant identity is elementary once the two definitions are aligned, but its value is organisational. It compresses three layers:

auxiliary determinant $\longrightarrow$ Jacobian rank $\longrightarrow$ smooth point avoiding one hypersurface.

For computation, the full determinant and its alpha-matrix implementation can be removed from the search criterion. For geometry, one may use any method that produces a reduced point away from the next coefficient. For arithmetic, a simple residue-class point can be lifted even when the original determinant vanishes modulo the chosen prime.

The three residues $g_{d+e-1,e}$, $J_{d,e}$ and $a_{d,e}$ may still be evaluated redundantly to catch software errors. They are not algebraically independent on the common-zero locus: equation (4.4) couples them exactly.

9.2 What remains difficult

Internal reproducibility is no longer the main bottleneck. The decisive open problem is structural: prove that at least one good affine point exists for all d in an unresolved column, or prove the stronger Conjecture 7.4. Three routes are now sharply separated:

1. prove uniform boundary coprimality and toric non-degeneracy;
2. exclude triple common zeros and singular intersections by recurrence or resultant identities;
3. construct a simple point asymptotically and certify a finite complement.

Failure of this particular smooth-point route at one d would not disprove the Polydegree Conjecture. Lewis–Perry–Straub Theorem 4 gives a sufficient condition (Lewis et al. 2019).

9.3 Significance and priority

The paper’s current contribution is a reusable theorem package: a transparent Jacobian interpretation, an exact syzygy, a sharper Hensel criterion and a curve-theoretic bridge. The finite $e = 3$ geometry identifies a precise frontier rather than another arbitrary cutoff. A uniform theorem would materially increase significance; a larger finite table would not.

Priority remains provisional. No inspected source states the present derivative–determinant–Euler chain, but the first identity is close to the multinomial reindexing already used in Lewis–Perry–Straub, and elementary observations are particularly vulnerable to unpublished or implicit prior use. Original-author contact and specialist review are therefore substantive next steps, not administrative decoration.

10 Limitations

1. The Jacobian identities and bridge theorems have written proofs and internal adversarial review, but no proof-assistant formalisation or external specialist validation.
2. The prior-art search is bounded. Authenticated MathSciNet, zbMATH and Scopus searches and author contact were not completed.
3. The full affine statement is checked only for $2 \leq d \leq 20$, and adjacent boundary coprimality only through $d = 200$. Uniform squarefreeness of each individual boundary factor does not imply adjacent coprimality. The finite bounds are evidence boundaries, not mathematical thresholds.
4. The 51 modular witnesses and their reimplementations originate within one research workflow. Different code paths do not constitute unaffiliated reproduction.
5. The manuscript inherits the correctness and interpretation of Lewis–Perry–Straub Theorem 4. It reproduces the hypotheses used but does not reprove their polynomial-automorphism construction.
6. No claim is made that the work currently meets a particular REF grade. Originality, significance and rigour require separate assessment, and publication infrastructure does not determine any of them.

11 Conclusion

The Lewis–Perry–Straub determinant is the Jacobian of their coefficient map. Weighted homogeneity turns that identity into an exact syzygy and a smooth-point reformulation of the specialisation criterion. The resulting finite-field certificate is smaller and strictly more permissive: it needs a simple common zero and a non-zero next coefficient, even when the prime divides $d + e - 1$. The same formulation permits a divisor on a smooth curve to produce the required point.

For $e = 3$, exact calculations show a notably strong finite pattern—the entire affine intersection is good through $d = 20$ —while the boundary factors are uniformly squarefree. These results isolate a clean uniform conjecture. The responsible next advance is a structural proof or counterexample, not a longer table. Until then, the general Jacobian and Hensel theorems stand as the paper’s proved contribution, while the all- d statement remains open.

Declarations

Data and code availability

All scripts, frozen witness data, exact JSON receipts, claim-status metadata and SHA-256 manifests used for this candidate are included in the accompanying reproducibility package. No DOI or public repository is claimed at this stage. Public URLs and immutable identifiers must be added only after a separately verified release.

Ethics statement

This is theoretical and computer-assisted mathematics. It involved no human participants, personal data, animals or clinical materials; institutional ethics approval was not applicable.

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AI and automated-tool disclosure

Automated research, symbolic-computation and language-model tools assisted literature triage, proof exploration, exact verification, adversarial review, drafting and package preparation under human direction. They are not listed as authors. The human author is responsible for the submitted claims, source selection, final text and release decision. Agreement among AI-assisted checks is not represented as independent human review.

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